

Problem

Using a single op amp, a capacitor, and resistors of 100 kΩ or less, design a circuit to implement

$$v_o = -2 \int_0^t v_i(\tau) d\tau$$

Assume $v_o = 0$ at $t = 0$.

Step-by-step solution

Step 1 of 2

The output voltage of the required circuit is

$$v_o = -2 \int_0^t v_i(t) dt \dots\dots (1)$$

Here, the output is proportional to the integral of the input signal.

An integrator is an op amp circuit whose output is proportional to the integral of the input signal.

Refer to the Figure 1, for op amp integrator circuit.

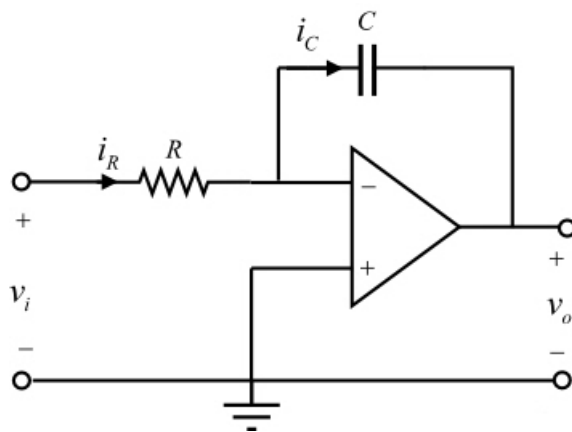


Figure 1

Write the formula for output voltage of the integrator.

$$v_o = -\frac{1}{RC} \int_0^t v_i(t) dt + v_o(0) \dots\dots (2)$$

Assume zero for $v_o(0)$.

Compare the two equations (1) and (2).

$$\frac{1}{RC} = 2$$

Assume 100 kΩ for R .

$$\frac{1}{100\text{k} \times C} = 2$$

$$C = 5 \times 10^{-6}$$

$$C = 5 \mu\text{F}$$

Step 2 of 2

Hence, the op amp circuit with the required specification is shown in Figure 2.

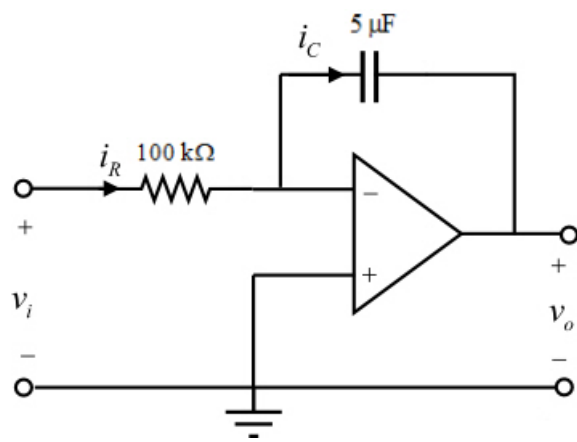


Figure 2